MedCheckMe App: A Medical Diagnostic Application

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Abstract

Lack of access to medical checkups in developing countries has resulted in a significant impact on health. The MedCheck App aims to address this issue with an efficient screening multi-purpose application for monitoring population's health. It will utilize new technology to provide easy access to low-cost medical checkups. It will utilize new technology to provide easy access to low-cost medical checkups. The MedCheck App will conduct four analyses: 1) Eyesight, 2) Hearing, 3) Skin, and 4) Heart rate, blood pressure and respiration. Results from these metrics will provide remote diagnosis on their health status. Using relevance of data, ease of measurement, accuracy and feasibility, and use of attachment, the tests were chosen to monitor the patient while being lost cost. The goal is to screen patients effectively to monitor and reduce the risk of non-communicable diseases in Haiti and Costa Rica.

Background

The Social ventures Foundation is a non-profit organization that supports social entrepreneurship. They use microfranchising to provide economic development and create jobs to help communities step out of poverty. The MedCheckMe App aims to address this issue with an efficient screening multi-purpose application for monitoring population's health. It will utilize new technology to provide easy access to low-cost medical checkups.

Prior Art

Many devices and apps and attachments exist in the realm of healthcare including telehealth apps, apps that utilize a phone's hardware and software, and apps that use external attachments. Some of these apps include Apple Health, MDLIVE, and Welltory. These apps are extremely useful, but many of them require a strong internet connection, expensive attachments, or continual use to be effective. MedCheckMe aims to take the best parts of these apps and combine them into one multi-functional app that can be used for medical check-ups in remote areas.

Methods

The MedCheckMe App provides the monitoring capabilities of:

1. Cardiology

Blood pressure and heart rate are important indicators of a patient's health and are essential to many diagnostic and therapeutic decisions [1]. One method for quantifying these parameters is Photoplethysmography (PPG). PPG is an optical technique used to detect volumetric changes in blood circulation and represent the data as a waveform.

The application employs a dual-faceted approach to obtain readings of a patient's heart/respiration rate, analyzing the frequency components of the measured PPG Waveform while simultaneously obtaining the rolling average of a series of momentary windows in order to generate a more robust estimate. These operations conduct themselves in unison with a series of functions monitoring for motion artifacts and/or filtering out ambient light interference, in addition to a rejecting data based on a quality assessment module which evaluates the signal-to-noise ratio.

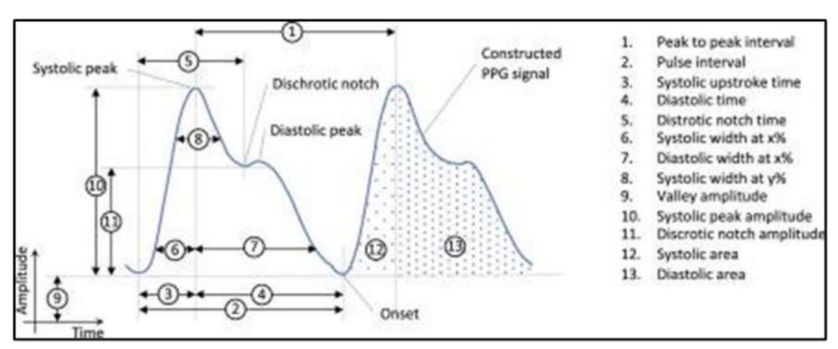

Figure 1. Representation of how to derive heart rate and blood pressure values from PPG signal.

To collect PPG data a video is recorded of a fingertip covering a camera and a light source. To ensure meaningful data is recorded for at least 90 seconds. From there data can be collected and processed as follows:

- The mean light intensity is acquired from the pixels in a selected region of interest.
- The data is then run through a series of band pass filters that remove frequencies out of range, noise, and wander.
- The resulting data represents a raw PPG signal and can be plotted against time to produce a waveform as seen in Figure 1.

To estimate blood pressure MedCheckMe will make use of a machine learning algorithm. Using important features such as systolic peak, diastolic peak, and dicrotic notch extracted from a raw PPG waveform a machine learning algorithm can produce an estimate for blood pressure by mapping the extracted features to waveforms within its training dataset and their associated projections and blood pressure values.

2. Audiology

Regular hearing assessments are important to detect hearing degradation and other hearing problems before they become too difficult to treat. Hearing tests become more imperative for populations at greater risk for hearing loss, such as elderly or industrial workers.

Hearing levels are evaluated using the Pure Tone test method. Payback of a range of frequencies (250-8000 Hz) at increasing decibels occur via earbuds. The patient responds which decibels per frequency they can hear. The lowest audible decibel is then recorded and displayed in an audiogram for data analysis and hearing level identification. This test method is repeated three times: one for both ears, the left and the right ear.

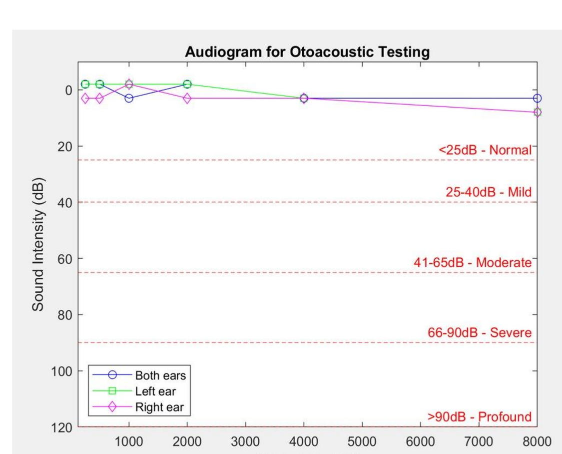

Figure 3. A representative audiogram from hearing test. Frequencies ranging from 0 to 8 KHz were evaluated for a female healthy patient (21 yo). Hearing levels are presented as Normal.

3. Optometry

Visual Acuity is a measurement of clarity and sharpness of one's vision and provides information for diagnosing and managing eye diseases. Visual acuity for MedCheckMe is measured using Tumbling E charts. Patient is shown letter E pointing in different orientations (up, down, left, right). Patient stands 2-3 meters away from the smartphone, covers one eye at a time, and indicates to the technician which direction they perceives the letter pointing at. Ishara color blind plates measure color deficiency of the congenital origin. Each plate held 75 cm from the patient one at a time. Out of 11 plates, 10 or more correct is normal vision, less than 7 is deficiency.

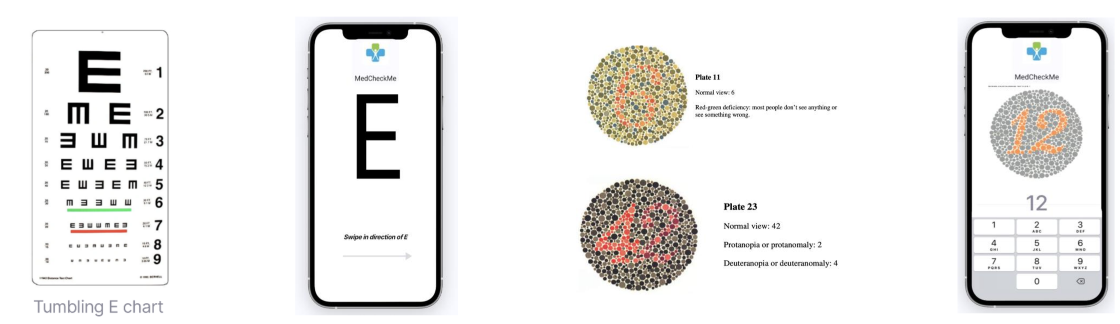

Figure 3. Visual Acuity Test on smartphone adapted from tumbling E chart (Left). The Ishihara Color Blind Plates (Right).

4. Dermatology

Dermoscopy is a method of visualizing skin lesions and classifying them by examining the lesion characteristics. Images taken with a low-cost dermoscopic solution attached to a phone's back camera undergo a series of transformations which seek to provide more detail about lesion characteristics than was initially seen. Diagnosis of these lesions will be done in the future using A.I, as access to a dermatologist is sparse within the rural areas that we seek to implement the MedCheck App. Machine Learning Algorithms are an effective tool in diagnosing skin lesions in the absence of a dermatologist by analyzing a database of images and characterizing qualities of different lesion types to come to an accurate assessment [2].

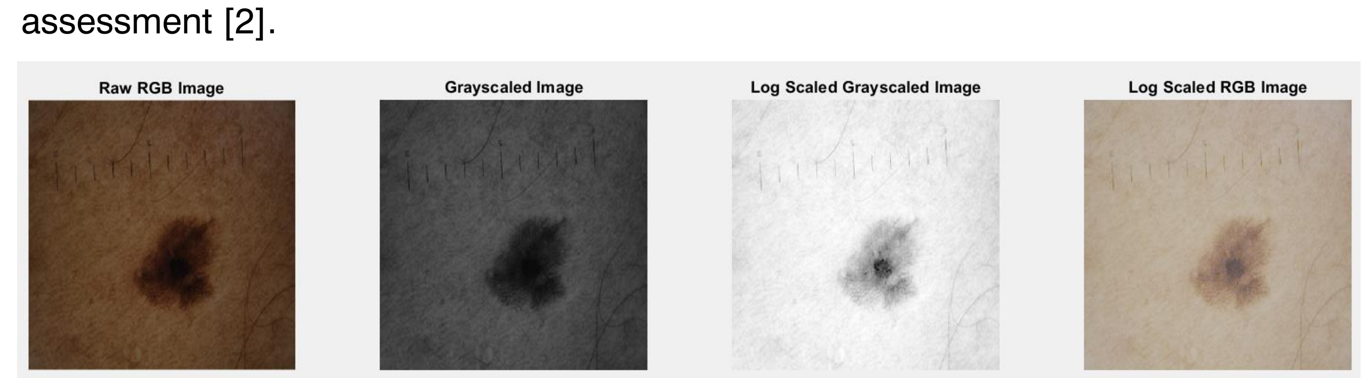

Figure 4. MATLAB proof of concept for filtering skin anomalies for dermoscopy applications.

5. Phone Selection

For the MedCheckMe platform, camera quality is critical for cardiology and dermatology. Storage, battery life, and cost are also important for overall efficacy. Various phone types were evaluated, with Google Pixel Phones scoring the highest

Criteria	Cost	Battery Life	Camera	Storage	Relevance	Score
Weight	35	15	25	10	10	
Samsung Galaxy S20	3.61	2.26	2	5	3.72	2.97
Samsung Galaxy S21 5G	2.63	2.53	4	5	3.72	3.17
Samsung Galaxy S22	2.49	2.51	4	5	3.72	3.12
Samsung Galaxy S23	0.85	4.40	5	5	3.72	3.08
ASUS ROG Phone 5S	2.82	3.94	3	5	3.72	3.20
ASUS ROG Phone 6	1.97	4.85	5	5	3.72	3.41
Samsung Galaxy a32	4.00	4.52	3	3	3.72	3.50
Samsung Galaxy A54	3.80	5.00	3	5	3.72	3.70
iPhone 8	4.85	1.90	1	3	0.62	2.60
iPhone X	4.79	1.85	2	3	0.62	2.81
iPhone 11	2.49	4.47	3	3	0.62	2.65
iPhone 12	1.84	3.97	4	3	0.62	2.60
iPhone 13	1.51	2.63	5	5	0.62	2.61
Pixel 7	2.16	3.26	5	5	3.72	3.37
Pixel 6	3.80	3.45	5	5	3.72	3.97
Pixel 5	4.92	3.79	3	5	3.72	3.91
Pixel 4	5.00	1.00	2	3	3.72	3.07

Figure 5. Phone trade study to determine model and OS.

Design Requirements

1. Diagnostic check-ups shall be kept below $1
2. App shall provide pertinent diagnostic tests conforming to regulatory standards
3. App shall not require constant internet connection
4. Patient information shall be kept confidential and secure

Integrating MedCheckMe

The prototype app was coded using Android Developer software, and tested on a virtual Google Pixel 5 emulator

A relational database was designed and tested using SQLite and Postgre SQL

The integrated MedCheckMe platform begins with an administrator providing diagnostic tests and questionnaires via the app. The patient information is then encrypted, deidentified, and stored in a postgre relational database using Google Cloud. The analysis of the diagnostic data is returned and de-encrypted using a biometric identifier. The resulting analysis, represented by a red, yellow, or green flag, reports to the patient the expected diagnosis (if present) and necessary next steps.

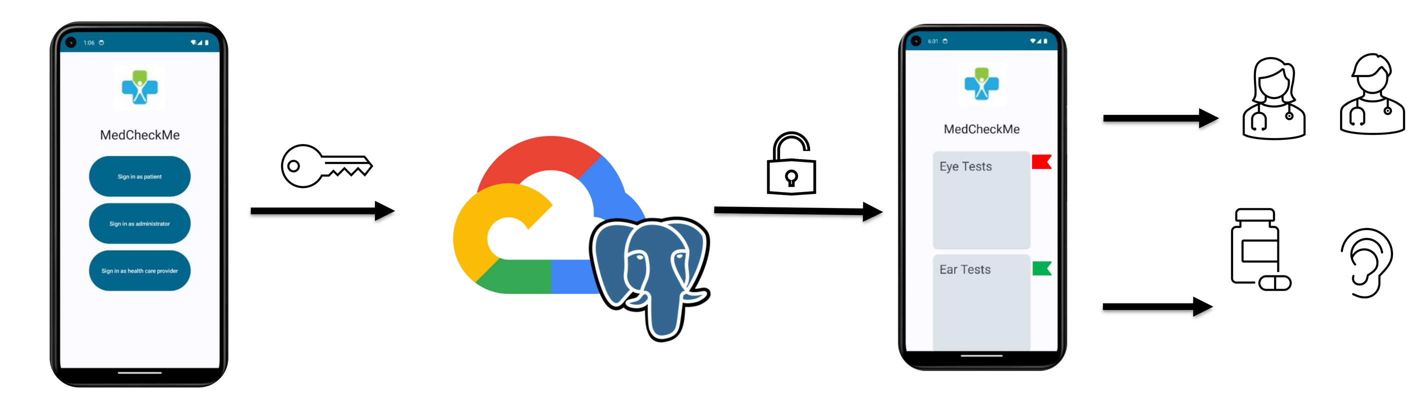

Figure 6. MedCheckMe Platform integration pathway from administration to analysis.

Future Work

Although the MedCheckMe proof of concept was successful, to provide a robust platform this team proposes the following future steps:

1. Perform "boots on the ground" testing of prototype platform in Haiti with Partners in Health Haiti.
2. Introduce and integrate further diagnostic tests, such as respiratory tests, inner ear exams, retinal scans, and more.
3. Adapt app to work on multiple OS and phone models.
4. Translate MedCheckMe platform for use in other countries pending successful proof of concepting in Haiti

Summary

- The MedCheckMe app provides diagnostic capabilities for cardiology, dermatology, optometry, and audiology.
- The platform is prototyped on a Google Pixel 5, utilizing android developer and google cloud platform software.
- The app design will continue beyond capstone for future utilization in Haiti

References

[1] https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2508710/#:~:text=Arterial%20blood%20pressure%20and%20heart,many%20diagnostic%20and%20therapeutic%20decisions.
[2] Machine Learning in Dermatology: Current Applications, Opportunities, and Limitations - PMC (nih.gov)

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